MIL-D-50998
7 December 1971

MILITARY SPECIFICATION

DETONATOR, STAB, M76

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing of a stab type delay detonator designated the M76.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-398 - RDX
MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-D-22264 - Powder, Ignition, Gasless

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosives; Sampling, Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1330

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9234579 - Detonator, Stab, Delay, M76
9258182 - Box, Set-up, Packing, Ammunition, Inner for Small Explosive Components
9258183 - Box, Set-up, Packing, Ammunition, Outer for Small Explosive Components
9258184 - Box, Packing, Ammunition for Small Explosive Components

PUBLICATIONS

U.S. ARMY MUNITIONS COMMAND

ET-9234579 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Assembly.-The detonator assembly shall comply with all requirements specified on Drawing (dwg.) 9234579 and with all requirements specified in applicable specifications.

3.2 Moisture content.-The moisture content of the following explosives at the loading station at the time of loading the detonators shall not exceed the percentage specified in Table I when tested as specified in 4.4.1.

TABLE I

Explosive	Percentage	Test Procedure
Primer mixture	0.30 maximum (max.)	4.4.1.1
Igniter mixture	0.30 max.	4.4.1.2
Lead azide	0.50 max.	4.4.1.3
RDX	0.50 max.	4.4.1.4
Delay Composition	0.50 max.	4.4.1.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

3.3 Functioning.-The detonator assembly shall not fail to fire on the first attempt and shall comply with the following requirements when conditioned at ambient temperature, cold temperature and water immersion.

3.3.1 Time.-The detonator assembly shall fire on the first attempt in not less than thirty (30) nor more than seventy (70) milliseconds when tested as specified in 4.4.2.1.

3.3.2 Output.-The detonator assembly shall fire on the first attempt under an applied energy of sixteen (16) inch ounces maximum (max.) and shall make a 0.567 inch minimum (min.) diameter perforation in a 0.131 inch min. thick lead disc with a 1/8 inch max. hole in its center when tested as specified in 4.4.2.2.

3.4 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.5 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that a statement of findings for the moisture content of each explosive shall be submitted with each lot submitted for acceptance.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation as specified herein. The first article shall consist of inert parts (see 4.2.2) and loaded detonators (see 4.2.3) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including certificates of conformance of materials. A sample containing known defects

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer shall also be submitted whenever a change occurs in manfacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inert parts.-A sample of twenty-five (25) each of every part shall be submitted in accordance with the provisions of 4.2.1, for inspection as specified in 4.2.2.1.

4.2.2.1 Inspections to be performed.-The parts will be subjected by the Government to any or all of the applicable examinations specified in 4.3.2 of this specification and any or all requirements of the applicable drawings.

4.2.2.2 Rejection.-If any part fails to comply with any of the requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon the failure on an item to comply with any of the requirements.

4.2.3 Loaded detonators.-The contractor shall submit a sample of forty-two (42) detonator assemblies to the Government for examination as specified in 4.3.2.6 and tested as specified in 4.2.3.1.

4.2.3.1 Tests to be performed.-Thirty-two (32) detonator assemblies shall be cold temperature conditioned and fired to determine time and output, and ten (10) detonators shall be water immersion conditioned and fired for time and output determination.

4.2.3.2 Rejection.-The loaded first article sample shall be rejected if any detonator fails to comply with a requirement. In the event of rejection, the Government reserves the right to terminate its inspection upon any failure of a detonator to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

only that quantity of units from which the sample was drawn and shall not be construed to represnet any prior of subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being aplied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded detonators shall contain:

a. Metal parts from not more than one lot interfix number from one supplier.

b. Primer mix PA-100, igniter mixture, or delay composition.-Each explosive from not more than one batch or blend of batches which results in homogeneity of each explosive used in loading a lot.

c. Lead azide of the same type from not more than two (2) lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformity controlled drying and operating conditions at the loading plant.

d. RDX from not more than one lot from one supplier.

4.3.2 Examination.-The examination listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and AQL of 0.65 percent for each minor defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Body (see dwg. 9234582 covering a detail of dwg. 9234579.)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Large outside diameter of body max	Gage	01001
102.	Diameter of counterbore, max	Gage	01002
103.	Diameter of hole	Gage	01003
104.	Small outside diameter	Gage	01004
105.	Depth of counterbore	Gage	01005
106.	Length of small outside diameter	Gage	01006
107.	Length from bottom of counterbore to small diameter end	Gage	01007
108.	Split, cut or crack	Visual	01008
Minor:	AQL 1.00 percent		
201.	Finish improper	Visual	01009
202.	Evidence of poor workmanship other than Major above (see 3.4)	Visual	01010

4.3.2.2 Disc, flash separator (see dwg. 9234581 covering a detail of dwg. 9234579.)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Diameter, min	Gage	02001
102.	Presence of holes	Visual	02002
103.	Diameter of holes, min	Gage	02003
104.	Burrs around holes	Visual	02004

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

Minor:	AQL 1.00 percent		
201.	Thickness	Gage	02005
202.	Evidence of poor workmanship other than Major above (see 3.4).	Visual	02006

4.3.2.3 Disc, closing (see dwg. 9255155 covering a detail of dwg. 9234579).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Diameter	Gage	03001
102.	Thickness	Gage	03002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	03003

4.3.2.4 Disc, detonator (see dwg. 9234580 covering a detail of dwg. 9234579).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Diameter	Gage	04001
102.	Thickness	Gage	04002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	04003

4.3.2.5 Assembly, prior to loading igniter powder and primer mixture (see dwg. 9234579).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

Major:	AQL 0.40 percent		
101.	Hole in flash separator disc missing	Visual	05001
102.	Closing disc loose or perforated	Visual Manual	05002
Minor:	AQL 1.00 percent		
201.	Closing disc wrinkled or folded	Visual	05003
202.	Crimp not full 360 degrees	Visual	05004

4.3.2.6 Assembly (see dwg. 9234579).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Body cracked	Visual	06001
2.	Presence of explosive on outside of body	Visual	06002
Major:	AQL 0.65 percent		
101.	Length	Gage	06003
102.	Diameter, max	Gage	06004
103.	Seal missing or color of seal incorrect	Visual	06005
Minor:	AQL 1.50 percent		
201.	Detonator disc crimp not full 360 degrees	Visual	06006
202.	Detonator disc wrinkled or folded	Visual	06007
203.	Evidence of poor workmanship other than Critical above (see 3.4)	Visual	06008

4.3.2.7 Box, set-up, inner, after loading and prior to closing slide cover (see dwg. 9258182).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Filler and cushion assembly missing	Visual	07001

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

2.	Filler and cushion assembly improperly assembled	Visual	07002
3.	Stiffner missing	Visual	07003
4.	Slide cover assembled reversed	Visual	07004
Major:	AQL 0.40 percent		
101.	Release paper missing	Visual	07005
Minor:	AQL 1.00 percent		
201.	Marking missing, incomplete or illegible	Visual	07006
202.	Evidence of poor workmanship (see 3.4)	Visual	07007

4.3.2.8 Box, set-up, outer, with inner set-up boxes assembled and prior to assembling top (see dwg. 9258183).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Red surfaces on inner set-up boxes do not alternate on same side of outer set-up boxes	Visual	08001
2.	Stiffner not facing one direction	Visual	08002
Major:	None defined		
Minor:	AQL 1.00 percent		
201.	Marking missing, incomplete or illegible	Visual	08003
202.	Evidence of poor workmanship (see 3.4)	Visual	08004

4.3.2.9 Box, packing, ammunition prior to assembling cover and strapping (see dwg. 9258184).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Bags containing set-up boxes damaged	Visual	09001

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

102.	Cushioning material incomplete or missing from around outer set-up boxes	Visual	09002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	09003

4.3.2.10 Box, packing, ammunition (see dwg. 9258184).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Strapping missing, broken, loose or improperly fastened	Visual-Manual	10001
102.	Box damaged	Visual	10002
103.	Marking missing, broken	Visual	10003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	10004

4.3.3 Testing.-Unless otherwise specified herein, the following tests shall be performed using MIL-STD-105 for sampling plan requirements.

4.3.3.1 Moisture content (see 3.2, Table I), Major Defects

TABLE II

Explosive	Code No.
Primer Mixture PA-100	11001
Igniter Mixture	11002
Lead Azide	11003
RDX	11004
Delay Composition	11005

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

The contractor shall provide adequate controls to insure that the explosives comply with the requirements. For verification, the contractor shall select a sample for test from that quantity of each explosive used in each eight hours of loading production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirements and loading has not begun, that quantity of explosive represented by the sample shall be rejected. If components have been loaded with explosive containing excessive moisture, the remaining unloaded explosive and the loaded components shall be rejected. The test procedures shall be as specified in 4.4.1.

4.3.3.2 Functioning after ambient, cold temperature and water immersion conditioning.

4.3.3.2.1 Sampling.-Four hundred and sixty-five (465) detonator assemblies shall be selected from each lot for test. The number of samples to be subjected to each test shall be as specified in Table III.

TABLE III

Test	Classification	Code No.	No. of Samples	Test Para.
Ambient temp. condition (see 3.3)	-------	----	315	4.4.2
Time (see 3.3.1)	Major	06009	---	4.4.2.1
Output (see 3.3.2)	Major	06010	---	4.4.2.2
Cold temp. condition (see 3.3)	-------	----	125	4.4.2
Time (see 3.3.1)	Major	06012	---	4.4.2.1
Output (see 3.3.2)	Major	06013	---	4.4.2.2
Water immersion (see 3.3)	-------	----	25	4.4.2
Time (see 3.3.1)	Major	06014	---	4.4.2.1
Output (see 3.3.2)	Major	06015	---	4.4.2.2

4.3.3.2.2 Rejection.-The lot shall be rejected if any of the following test results occur:

a. Ambient:
   1. Time.-Six (6) or more detonator assemblies fail to comply with the requirement.
   2. Output.-Two (2) or more detonator assemblies fail to comply with the requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

b. Cold Temperature:
   1. Time.-Five (5) or more detonator assemblies fail to comply with the requirement.
   2. Output.-Two (2) or more detonator assemblies fail to comply with the requirement.

c. Water Immersion.-Two (2) or more detonator assemblies fail to comply with the requirements.

4.3.3.3 Composition or primer mixture PA-100, igniter mixture, and delay composition (see dwg. 9234579).-The composition of each batch or blend of batches of each explosive shall be determined at the time of loading. A sample of sufficient size to perform the applicable test specified in 4.4.3 shall be selected. If the composition of the sample fails to comply with the drawing requirements, the quantity of explosive represented by the sample shall be rejected. Any detonator loaded with the non-conforming explosive shall be rejected.

4.3.4 Inspection equipment.-Equipment Tabulation Number ET-9234579 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test Methods and Procedures

4.4.1 Moisture content

4.4.1.1 Primer mixture PA-100.-Transfer a known value of between 0.4 to 0.7 gram of the sample to a tared weighing dish approximately 55 millimeters (mm) in a diameter and 30 mm in height.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

Place the dish and contents in an oven maintained at 55 to 65 degrees Centigrade (C.) or in a desiccator containing anhydrous calcium chloride until constant weight is obtained. Calculate the loss in weight as the percentage of moisture.

4.4.1.2 Igniter mixture.-Gasless igniter.-The moisture content of the gasless igniter shall be determined in accordance with MIL-P-22264.

4.4.1.3 Lead azide.-The moisture content of the lead azide shall be determine in accordance with Method 101.5 of MIL-STD-650 except that a two (2) gram sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene.

4.4.1.4 RDX.-The moisture content of the RDX shall be determined in accordance with the method in MIL-STD-398.

4.4.1.5 Delay composition.-Transfer an accurately weighed portion of approximately two (2) grams of the sample to a tared shallow receptacle and dry in an oven for 2 hours at 100 plus or minus 5 degrees C. Cool in a desiccator and weigh. Calculate the loss in weight of the receptacle and contents to percent moisture in the sample.

4.4.2 Functioning.-Prior to testing, the applicable quantity of detonator assemblies (see Table III) shall be conditioned as follows:

a. Ambient temperature.-The detonators shall be conditioned at seventy (70) plus or minus ten (10) degrees Fahrenheit (F.) for two (2) hours min.

b. Cold temperature.-The detonators shall be conditioned in the test fixture for twenty-four (24) plus or minus four (4) hours at minus sixty-five (65) degrees F. The detonators shall be fired within two (2) minutes after removal from cold chamber.

c. Water immersion.-The detonator shall be immersed for forty-eight (48) hours min. at a two (2) to three (3) inch depth of water maintained at a temperature of seventy (70) plus or minus ten (10) degrees.

4.4.2.1 Time.-After conditioning the detonator shall be assembled in a suitable test fixture so that the firing pin will strike the sensitive end of the detonator. A steel ball weighing two (2) ounces max. shall be dropped from a height of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

eight (8) inches max. on the top of the firing pin. The elapsed time between the ball striking the firing pin and the explosion shall be measured to determine compliance with the requirement. Detonators that fail to fire shall be recorded.

4.4.2.2 Output.-After conditioning, the detonators shall be assembled in the test fixture specified in 4.3.4 and fired as specified in 4.4.2.1. The diameter of each lead disc perforation shall be measured to determine compliance with the requirements.

4.4.3 Composition of primer mixture PA-100 , igniter mixture and delay composition shall be determined as follows:

4.4.3.1 Primer mixture PA-100

4.4.3.1.1 Potassium chlorate

4.4.3.1.1.1 Preparation of solutions

4.4.3.1.1.1 Solution No. 1.-To approximately 800 ml of distilled water in a 1-liter amber glass reagent bottle, 40 gm of potassium sulfocyanate shall be added. The reagent bottle and contents shall be sealed with a rubber stopper and shaken until the solid material has been dissolved. To this solution, 1.0 gm of lead thiocyanate, 1.0 gm of antimony sulfide, and 1.0 gm of lead azide shall be added. Again the reagent bottle shall be stoppered, and the contents vigorously shaken for several minutes, then allowed to stand over-night. Portions of the saturated solution shall be filtered before use.

4.4.3.1.1.1. Solution No. 2.-To approximately 400 ml of distilled water in a 1/2-liter amber glass reagent bottle, 1.0 gm of lead thiocyanate, 1.0 gm of antimony sulfide, and 1.0 gm of lead azide shall be added. The reagent bottle shall be sealed with a rubber stopper, and the contents vigorously shaken for several minutes, then allowed to stand overnight. Portions of the saturated solution shall be filtered before use.

4.4.3.1.1.2 Procedure.-A portion of approximately 1.0 gm of the sample shall be accurately weighed directly into a tared 30-ml sintered glass crucible of medium porosity. From a burette, 3.0 ml of Solution No. 1 shall be added to the crucible. The solution shall be swirled for exactly 1 minute, care being taken that no liquid be spilled, then aspirated with suction, and the bottom of the crucible wiped dry with tissue paper. The washing shall be continued, using one more 3.0 ml portion, two 2.0 ml portions, and two 1.0 ml portions, making a total of 12.0 ml in all. The crucible shall be removed, and the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

bottom of the crucible, the funnel, the adapter, and the suction flask washed with water. The test tube shall be placed in the flask and the stem of the filtering funnel inserted into it. As above, the crucible and contents shall be washed with one 1.0 ml portion of Solution No. 1. The solution shall be aspirated with suction, and the filtrate caught in the test tube. The test tube shall be removed, 1.0 ml of distilled water added, then tested for the chlorate ion, as follows:

The tube shall be inclined at an angle of about 45 degrees. About 0.5 ml of a solution containing 1.0 gm of diphenylamine in 100 gm of concentrated sulfuric acid shall be run down the side of the tube from a dropper so that there will be two distinct layers. If an appreciable amount of chlorate ion is present, a distinct blue ring shall be visible at the border of the two layers. On shaking the tube, the ring disappears, but the contents assume a blue coloration, which shall last several seconds, depending on the amount of chlorate ion present. If only a trace of chlorate ion is present, the blue ring may not form, but upon shaking the test tube, a slight bluish coloration shall appear momentarily throughout the solution, and last for only a fraction of a second. If a definite blue ring forms, the washing shall be repeated as before, using one 1.0 ml portion of Solution No. 1, and finally a 0.5 ml portion of the same solution, as needed, depending upon the outcome of the tests for the chlorate ion. (Usually a total of 14.0 or 14.5 ml of Solution No. 1 shall be sufficient.) After the removal of potassium chlorate has been completed, the inside of the crucible shall be washed with 1.0 ml of Solution No. 2, and aspirated immediately. The bottom of the crucible shall be wiped dry with tissue paper, and the crucible dried in an oven at 80 degrees plus or minus 1 degree C. for 30 minutes, cooled in a desiccator, and weighed. The percentage of potassium chlorate shall be calculated as follows:

$$\text{Potassium chlorate, percent} = \frac{B \times 100}{W}$$

where:

B = Loss in weight of the contents of the crucible.
W = Weight of the sample on a moisture-free basis.

4.4.3.1.2 Antimony sulfide

4.4.3.1.2.1 Preparation of solution of ferric ammonium sulfate.-To approximately 120 ml of distilled water in a 1/2-liter reagent bottle, 125 gm of ferric ammonium sulfate shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

be added. The reagent bottle shall be stoppered, vigorously shaken for several minutes, and then allowed to stand overnight. Portions of the saturated solution shall be filtered before use.

4.4.3.1.2.2 Procedure.-A portion of approximately 1 gm of the sample shall be accurately weighed directly into a tared 30-ml sintered glass crucible of medium porosity. A 5.0 ml portion of 1 percent nitric acid shall be added, and the crucible constantly swirled for 1 minute, in order that all the ingredients of the powder except antimony sulfide be dissolved. Care shall be taken that none of the liquid be spilled. The mixture shall be filtered by applying suction, employing a vacuum flask. The crucible shall be removed form the adapter, and the bottom wiped dry with tissue paper. The washing operation shall be repeated, using 5.0 ml of 1 percent nitric acid each time, and the contents swirled for exactly 1 minute. After a total of 40 ml of 1 percent nitric acid has been used, the crucible shall be removed, and the bottom of it washed with water. The funnel, the funnel adapter, and the vacuum flask shall be washed with water, and the filtrate and washings discarded. About 1 ml of a saturated solution of ferric ammonium sulfate shall be placed in the flask. The contents of the crucible shall be treated as before with 5.0 ml of 1 percent nitric acid, and filtered by means of suction into the flask. If the liquid in the flask turns red, due to the presence of the sulfocyanate ion in the filtrate, the washing of the contents in the crucible shall be repeated with additional 5.0 ml portions of 1 percent nitric acid, until a negative test for the sulfocyanate ion has been affected. (Usally a total of 50 ml shall be sufficient.) The crucible and contents shall be rinsed with alcohol, and then with ether. The solution shall be aspirated until the odor of ether can no longer be detected. The crucible and contents shall be dried in an oven at 80 degrees plus or minus 1 degree C. for 30 minutes, cooled in a desiccator, and weighed. The percentage of antimony sulfide shall be calculated as follows:

$$\text{Antimony sulfide, percent} = \frac{A \times 100}{B}$$

where:

A = Weight of residue in the crucible.
B = Weight of sample on a moisture-free basis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

4.4.3.1.3 Lead azide.-The lead azide content shall be determined using the (Eudiometer) method and apparatus in MIL-STD-650 Method No. 407.1.

4.4.3.1.4 Lead sulfocyanate.-The percentage of lead sulfocyanate in the sample shall be calcuated on a moisture free basis by subtracting from 100 the sum of the percentages of potassium chlorate, antimony sulfide, and lead azide.

4.4.3.2 Igniter mixture.-The composition of the igniter mixture shall be determined in accordance with MIL-P-22264.

4.4.3.3 Delay composition.-The composition of the delay composition shall be determined in accordance with Method 404.1 of MIL-STD-650.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-Packaging shall be as specified on dwgs. 9258182 and 9258183.

5.2 Packing

5.2.1 Level C.-Packing shall be as specified on dwg. 9258184.

5.3 Marking.-Markings of all packaging and packing shall comply with dwgs. 9258182, 9258183 and 9258184.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspection herein are to facilitate future

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

assigned to the inspection herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment
MU - Army design, mandatory for use

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classification: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

designs accordingly. The contractor may. however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procudure provided such format reflects the detail and information specified above, subject to the followign controls:

All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reporducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal. ATTN: SMUPA-QA-A-M. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of test data.-In addition to the normal distribution of records, when the detonator is procured by the U.S. Army Munitions Command one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-M, and SMUPA-AD-F-E-3.

Custodian:
Army MU

Preparing activity:
Army-MU

Project Number: 1330-A-010

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.